Raj Gandhi

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RESEARCH INTERESTS

I am interested in applying formal group laws and tools from generalized cohomology theories to study Schubert calculus, the geometry of matroids, and torus-equivariant degenerations.

EDUCATION

Cornell University — Ithaca, NY, USA
Ph.D. Candidate in Mathematics — *Aug 2021 – Current*
Advisor: Allen Knutson
Expected graduation: August 2027

University of Ottawa — Ottawa, ON, Canada
M.Sc. Mathematics — *Aug 2019 – Aug 2021*
Thesis title: *Oriented cohomology rings of the semisimple linear algebraic groups of ranks 1 and 2.*
Advisor: Alistair Savage and Kirill Zainoulline

University of Ottawa — Ottawa, ON, Canada
B.Sc. Physics-Mathematics — *Aug 2015 – May 2019*

AWARDS AND HONOURS

External awards

- NSERC Canada Graduate Scholarship, Doctoral (Declined) — *2021-2024*
- NSERC Postgraduate Scholarship, Doctoral (Accepted) — *2021-2024*
- Ontario Graduate Scholarship (Accepted) — *2020-2021*
- NSERC Canada Graduate Scholarship, M.Sc (Accepted) — *2019-2020*
- Ontario Graduate Scholarship (Declined) — *2019-2020*
- NSERC Undergraduate Student Research Award (Accepted) — *2017, 2018, 2019*

Internal awards

- Cornell Admission Fellowship. Accepted. (Awarded in 2021 and postponed) — *2024-2025*
- University of Ottawa Excellence Scholarship, M.Sc. Accepted twice — *2019-2021*
- University of Ottawa, Department of Mathematics and Statistics, Student Paper Award — *2018*
  (Awarded for paper titled "Decomposing Frobenius Heisenberg Categories")

RESEARCH

Preprints

1. (Joint with Gabe Udell). A cubulation of the associahedron using a torus-equivariant degeneration of the brick variety. Preprint (2026). arxiv.org/abs/2610.23004.
2. A technique for computing oriented cohomology rings of semisimple algebraic groups. Preprint (2026). arxiv.org/abs/2607.07900.
3. (Joint with Ethan Partida). Algebraic cobordism rings of wonderful varieties and matroids. Preprint (2026). arxiv.org/abs/2606.12645.
4. Motivic Segre classes of Schubert cells and the connective formal group law. Preprint (2026). arxiv.org/abs/2605.26556.

Publications

1. The formal affine Demazure algebra and real finite reflection groups. *Algebras and Representation Theory.* (2023). Vol. 26. No. 5. link.springer.com/article/10.1007/s10468-022-10158-x.
2. (Joint with Alistair Savage and Kirill Zainoulline). Diagrammatics for F_4. *Bulletin of the London Mathematical Society.* (2023). Vol. 55. No. 1. 1. londmathsoc.onlinelibrary.wiley.com/doi/10.1112/blms.12733.
3. Decomposing Frobenius Heisenberg categories. *Journal of Algebra and its Applications.* (2020). Vol. 19. No. 5. doi.org/10.1142/S0219498820500942.

Refereed Conference Proceedings

1. (Joint with Ethan Partida). Algebraic cobordism rings of wonderful varieties and matroids. *Séminaire Lotharingien de Combinatoire*. Issue 95B. To appear. sites.math.washington.edu/fpsac2026/public/abstracts/gandhi.pdf
2. Deforming the motivic Segre classes of Schubert cells in the Grassmannian. *Séminaire Lotharingien de Combinatoire.* 2025. Issue 93B. https://www.mat.univie.ac.at/~slc/wpapers/FPSAC2025/12.pdf
3. (Joint with Gabe Udell). A degeneration of the brick variety and a mixed subdivision of the associahedron into cubes. 2025. Issue 93B. https://www.mat.univie.ac.at/~slc/wpapers/FPSAC2025/91.pdf.

M.Sc. Thesis

Oriented cohomology rings of the semisimple linear algebraic groups of ranks 1 and 2. 2021.University of Ottawa library. *ruor. uottawa.ca/handle/10393/42566.*

INVITED AND CONTRIBUTED TALKS

- Algebraic cobordism rings of wonderful varieties and matroids *Oct 16, 2026*
  Combinatorics Seminar. University of Michigan.
- Algebraic cobordism rings of wonderful varieties and matroids *Oct 9, 2026*
  Algebra Seminar. Cornell University.

- Algebraic cobordism rings of wonderful varieties and matroids *May 2, 2026*
  Algebraic Geometry Northeastern Series (AGNES). Rutgers University (5-minute talk)
- Motivic Segre classes of Schubert cells and the connective formal group law *Feb 16, 2026*
  Informal Mathematical Physics Seminar. Columbia University
- Motivic Segre classes of Schubert cells and the connective formal group law *Feb 12, 2026*
  Geometry, Physics, and Representation Theory Seminar. Northeastern University
- Motivic Segre classes of Schubert cells and the connective formal group law *Jan 29, 2026*
  Algebra/Topology Seminar. University of Albany
- Deforming the motivic Segre classes of Schubert cells in the Grassmannian *Jun 26, 2025*
  Algebra Seminar. University of Ottawa
- Deforming the motivic Segre classes of Schubert cells in the Grassmannian *May 1, 2025*
  Combinatorics Seminar. University of Toronto
- Degenerating brick manifolds and subdividing the associahedron *Mar 17, 2024*
  Graduate Student Combinatorics Conference (GSCC). Carnegie Mellon University
- The formal group ring and real finite reflection groups *June 3, 2021*
  Algebra and Geometry of Homogeneous Spaces. University of Ottawa

POSTER PRESENTATIONS

- Algebraic cobordism rings of wonderful varieties and matroids *Jul 14, 2026*
  Formal Power Series and Algebraic Combinatorics (FPSAC). University of Washington
- Motivic Segre classes of Schubert cells and the connective formal group law *May 27, 2026*
  Affine Combinatorics and Quantum K-theory. Ohio State University
- Deforming the motivic Segre classes of Schubert cells in the Grassmannian *Jan 24, 2026*
  Combinatorial Algebra Meets Algebraic Combinatorics (CAAC). Dalhousie University
- Deforming the motivic Segre classes of Schubert cells in the Grassmannian *Nov 22, 2025*
  Algebraic Geometry Northeastern Series (AGNES). University of Massachusetts, Amherst
- Deforming the motivic Segre classes of Schubert cells in the Grassmannian *Jul 21, 2025*
  Formal Power Series and Algebraic Combinatorics (FPSAC). Hokkaido University, Japan
- Deforming the motivic Segre classes of Schubert cells in the Grassmannian *Mar 22, 2025*
  Mid-Atlantic Algebra, Geometry, and Combinatorics (MAAGC) Workshop. CUNY Graduate Center, NY
- Generalized cohomology rings of rank 2 roots systems *Jun 26, 2024*
  Encuentro Colombiano de Cominatoria (ECCO). Universidad del Cauca, Colombia
- The ring of twisted differential operators of reflection group $I_2(5)$ *2016*
  UROP Symposium. University of Ottawa. ruor.uottawa.ca/handle/10393/36283

EXPOSITORY TALKS

- Nakajima quiver varieties in type A. *Oct 14, 2026*
  Puzzles seminar. University of Michigan.

- Linear algebra through quiver representations. *Feb 20, 2024*
  Olivetti Club. Cornell University
- String calculus: a calculus with a twist! *May 2, 2023*
  Olivetti Club. Cornell University

TEACHIING EXPERIENCE

Cornell University **Ithaca, NY, USA**

Instructorships

- MATH 1110: Calculus I *Spring 2026*

Teaching Assistantships

- MATH 1110: Calculus I. Grader and recitation instructor *Fall 2026*
- MATH 1920: Multivariable Calculus for Engineers. Grader and recitation instructor *Fall 2025*

University of Ottawa **Ottawa, ON, Canada**

Teaching Assistantships

- MAT 1320X: Calculus I. Recitation instructor *Summer 2021*
- MAT 2355: Introduction of Geometry. Grader *Winter 2021*
- MAT 1348: Discrete Mathematics for Computing. Grader and recitation instructor *Winter 2021*
- MAT 2141: Honours Linear Algebra. Grader *Fall 2020*
- MAT 1341: Introduction to Linear Algebra. Grader and recitation instructor *Fall 2020*
- MAT 1332: Calculus II for the Life Sciences. Grader and recitation instructor *Fall 2020*
- MAT 1300X: Mathematical Methods I. Recitation instructor *Summer 2020*
- MAT 1320. Calculus I. Grader and recitation instructor *Winter 2020*
- MAT 1348: Discrete Mathematics for Computing. Grader and recitation instructor *Winter 2020*
- MAT 1362: Mathematics Reasoning and Proofs. Grader *Winter 2020*
- MAT 1362: Mathematical Reasoning and Proofs. Grader and recitation instructor *Fall 2019*

OUTREACH AND SERVICE

Most of my outreach material can be openly accessed through this link: drive

Ithaca High School (IHS) Mathematics Seminar

Goal: Develop and teach a 16-lecture course to Ithaca High School students and then guide a subset of them in a learning project.

Instructor

- Course title: ***Tropical geometry*** *Jan-Mar 2024*
  Mentee(s): Benjamin Wahl and Jessica Swart
  Project: Create "tropical geometry visualization" website
- Course title: ***Partially ordered sets*** *Jan-Mar 2024*
  Mentee(s): Taran Knutson
  Project: read the following paper of Fomin-Britz: arxiv.org/pdf/math/9912126
  Mentee(s): Niko Ross, Will Biloski, Will Cawley, Teddy Greenwald-Schaye, Jay Hirsh

Project: Learn various graph theory concepts

- Course title: ***Young tableaux*** *Nov-Dec 2024*
  Mentee(s): Divya Bhojraj
  Project: Implement RSK algorithm in Python

Cornell Prison Education Program (CPEP)

- Instructorship course: MATH 112: Contemporary Mathematics *Jan-Feb 2025*
  Location: Cayuga Community College, Auburn Correctional Facility
- Tutor course: MATH 112: Contemporary Mathematics *Fall 2024*
  Location: Cayuga Community College, Auburn Correctional Facility

Mentor at Canada/USA Mathcamp *Jun-Aug 2023*

Served as a mentor for the campers, and taught mini courses, with the following titles:

1. ***Knot Invariants***
2. ***Representation theory of the symmetric groups***
3. ***Linear algebra through knots***
4. ***Quiver representations II***
5. ***Introduction to Schubert calculus***

GRASSHOPR AT CORNELL

Goal: Develop and teach a 3-5-lesson long mini-course for K-12 students

- (Joint with Anna Barth). Course title: ***Cryptography*** *Mar 2025*
  Location: Lehman Alternative Community School, Ithaca NY
  Audience: 6th-8th graders
- Course title: ***Graph theory*** *Mar 2023*
  Location: Lehman Alternative Community School, Ithaca NY
  Audience: 6th-8th graders
- (Joint with Aria Beaupré). Course title: ***Voting theory*** *Mar 2022*
  Location: Namasté Montessori School, Ithaca NY
  Audience: 6th-8th graders

Directed Reading Program

Goal: Read a mathematics textbook alongside an undergraduate student, guiding them towards understanding of the material.

Cornell University

- Textbook: *Matroids: A Geometric Introduction,* by Gary Gordan and Jennifer McNulty *Fall 2025*
  Mentees: Idhant Gode and Nir Pathak
- Textbook: *The Knot Book*, by Colin C. Adams *Spring 2023*
  Mentee: Ronan Everett.

- Textbook: *Arithmetic of Elliptic Curves,* by Joseph H. Silverman *Fall 2022*
  Mentee: Ronan Everett

Seminars Organized

- "What is…?" seminar. Cornell University *Fall 2022*

CONFERENCES AND WORKSHOPS ATTENDED

- **Formal Power Series and Algebraic Combinatorics (FPSAC)** *Jul 13-17, 2026*
  University of Washington, WA
- **Affine Combinatorics and Quantum K-theory** *May 26-29, 2026*
  Ohio State University, OH
- **Algebraic Geometry Northeastern Series (AGNES)** *May 1-3, 2026*
  Rutgers University, NJ
- **Combinatorial Algebra Meets Algebraic Combinatorics (CAAC)** *Jan 23-35, 2026*
  Dalhousie University, NS
- **Algebraic Geometry Northeastern Series (AGNES)** *Nov 21-23, 2025*
  University of Massachusetts, Amherst, MA
- **Workshop on Algebraic and Representation Theory (WARTHOG)** *Aug 18-22, 2025*
  University of Oregon, OR
- **Formal Power Series and Algebraic Combinatorics** *Jul 21-25, 2025*
  Hokkaido University, Japan
- **Mid-Atlantic Algebra, Geometry, and Combinatorics (MAAGC) Workshop** *Mar 21-22, 2025*
  CUNY Graduate Center, NY
- **Apprenticeship Program in Commutative Algebra** *Jan 20-24, 2025*
  Field's Institute, ON
- **Formal Power Series and Algebraic Combinatorics (FPSAC)** *Jul 22-26, 2024*
  Ruhr-Universität Bochum, Germany
- **Summer School in Algebraic Combinatorics** *Jul 15-19, 2024*
  Max Planck Institute, Leipzig, Germany
- **Encuentro Colombiano de Combinatoria (ECCO)** *Jun 17-28, 2024*
  Universidad del Cauca, Colombia
- **Graduate Student Combinatorics Conference (GSCC)** *Mar 15-17, 2024*
  Carnegie Mellon University, PA
- **Schubert Summer School** *Jun 12-22, 2023*
  University of Illinois at Urbana-Champaign, IL
- **Algebraic Geometry Northeastern Series (AGNES)** *Apr 28-30, 2023*
  Stony Brook University, NY
- **Route 81**. Queen's University, ON *Sep 24, 2022*
- **The Los Angeles Workshop on Representation and Geometry** *Jun 6-10, 2022*
  University of Southern California, CA

- **Cornell Topology Festival**. Cornell University, NY *May 6-8, 2022*
- **Derived categories and moduli spaces**. Cornell University, NY *Apr 29-May 1, 2022*